Introduction

Key Business Problem

Our Real Estate group, MSBA Realty, is facing the key business problem of predicting accurate house sales prices in Ames, Iowa to better tailor our listing prices. In this business problem, we need to predict accurate house sales prices using property and home attributes available from assessments and historical housing data. Predicting accurate prices allows us to better tailor our listings to prevent overpricing. This helps reduce overpricing risks including extended time on the market, decreased buyer interest, and losses through price reductions. In the past, we have noticed that our pricing estimates tend to be higher than what buyers are willing to pay, so by analyzing historical data for homes in Ames and creating a model to better predict an appropriate price, we will be able to improve house sales times and revenue for our team.

Accurate pricing impacts how long houses sit on the market, future negotiation, and buyer trust. Our end goal is to create a data driven framework that helps our agents make informed pricing recommendations and keep up MSBA Realty's competitive advantage. The decision we are facing is how to price houses we will sell on the market, and what features are important to determine the price we set for a house.

Overview of Our Conclusions

Through our modeling approach using LASSO to select key features to improve upon our baseline linear regression model, we found that the most impactful features on housing price were home **square footage of above ground living area** ("GrLivingArea"), overall **material and finish quality** ("OverallQual"), the **size of a garage** in car capacity ("GarageCars"), the **year the house was built** ("YearBuilt"), and the **overall house condition** ("OverallCond"). These findings are important to our approach of determining house prices in Ames because we are able to use our knowledge of these features through appraisals (touring the house to evaluate it, or obtaining data about the features of a house from another source) to estimate house prices before we put them on the market for sale. This will allow us to have more accurate prices and therefore sell houses quicker by matching the customer willingness to pay.

Data

To inform our analysis, we used a dataset composed of historical information about houses in Ames, Iowa. This dataset has 79 explanatory variables describing aspects of residences such as lot area, year built, and overall condition. These features reflect real world factors that would influence housing prices that are available using previous appraisal data or touring houses before listing.

Data Cleaning and Transformation

The dataset we used for our analysis was loaded into two different files, training and test data, and read into Pandas dataframes in Google Colab. We then moved on to Data Cleaning and handling the columns with a large amount of null values. We dropped Alley, PoolQC, MasVnrType, FireplaceQu, Fence, and MiscFeature. For columns that represent missing values,

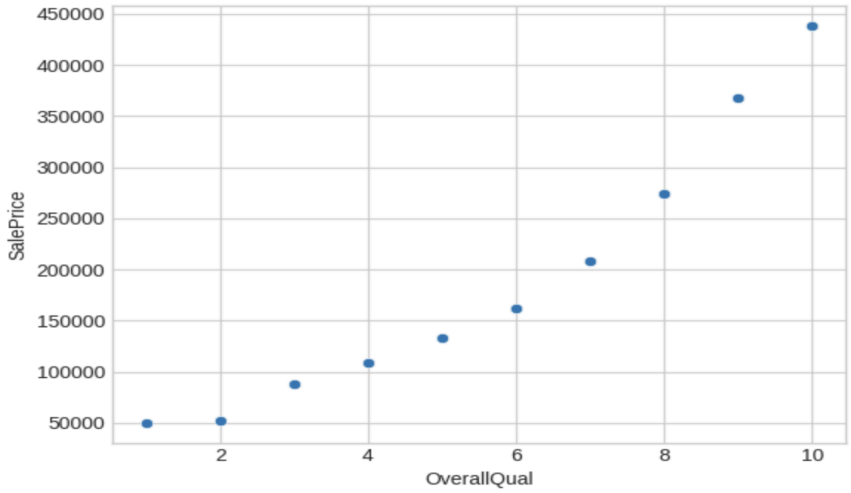
we evaluated what the missing values means. If the missing value meant that the feature does not exist, we replaced the null values with zero. This included columns like GarageType, GarageYrBlt, GarageFinish etc. For the rest of categorical variables, we filled them with the mode for the column to make sure that the unknown but applicable nature of the values were captured. For columns with numerical values BsmtUnfSF, GarageArea and LotFrontage, we filled them with mean values from the respective column to eliminate all possible null values. After cleaning the data, we split it back into test and training sets.

Feature Engineering

We attempted to simplify the dataset for example by dropping columns; BsmtFinSF, BsmtFinSF2, BsmtUnfSF to just keep the TotalBsmtSF column and by adding BsmtFullBath and BsmtHalfBath to one column for basement bathrooms. However, these changes did not impact our results when we tested different models. Therefore, we didn't engineer new features and let the model select the best ones. In the future, we would consider engineering the feature "HouseAge" by taking the year the house was sold and subtracting the year the house was built.

Exploratory Data Analysis

To conduct exploratory data analysis, we first looked at the distribution of sale prices. The sale prices were heavily right skewed, therefore we used a log transformation to make sure the model would perform better. We also explored the relationship between neighbourhoods and sale price. We found that the interquartile ranges and medians for neighbourhoods vary and data is dispersed for certain neighbourhoods more than others, however several neighbourhoods had very similar data distribution. We also looked at a scatterplot between OverallQual and their associated mean SalePrice. We can see there's a very strong relationship, indicating this feature might be a strong predictor.



Model

To effectively model the impact of housing features on predicted sales price, we compared two models: a baseline OLS linear regression model incorporating all of the housing features we examined (233 features including our dummy variables) and a refined model using LASSO

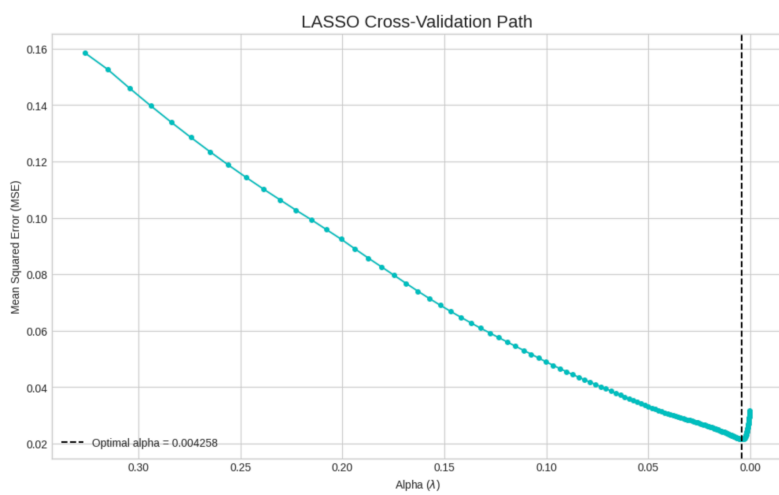
regression to shrink and select key features that had the greatest impact on our predictions. We decided to compare the two models to evaluate which one was more effective in prediction accuracy on test data. We believe that the LASSO Regression model will be more effective because it simplifies our complex model and reduces the risk of overfitting to the training data by performing automatic feature selection and reducing the coefficient or irrelevant features to 0.

Benchmark Model: Ordinary Least Squares (OLS) Regression

We used Ordinary Least Squares (OLS) regression as our baseline model. OLS is a simple and easily understandable linear model that minimizes the sum of squared differences between the observed and predicted values for housing prices. This provided a straightforward starting point to understand the basic linear relationships in the data and benchmark the performance of more refined models like LASSO (introducing regularization), allowing us to see if there was improvement in key metrics such as RMSE (Root Mean Squared Error) by using a different approach compared to OLS.

Proposed Improvement Model: LASSO Regression

We believed that the LASSO regression model would perform better due to the fact that our dataset has a relatively large number of features (233 after coding for dummy variables), therefore some of these features may not truly contribute to predicting the sale price. Our OLS linear regression model used all features, which could lead to overfitting. LASSO (Least Absolute Shrinkage and Selection Operator) adds a penalty term to our linear regression based on the absolute values of the coefficients, which encourages the model to shrink some coefficients towards or to zero. This effectively performs automatic feature selection by removing irrelevant features by setting their coefficients to exactly zero. In our improved model, LASSO selected 87 out of 233 features, which simplified the model and reduced the risk of overfitting by becoming less sensitive to the noise in the training data.



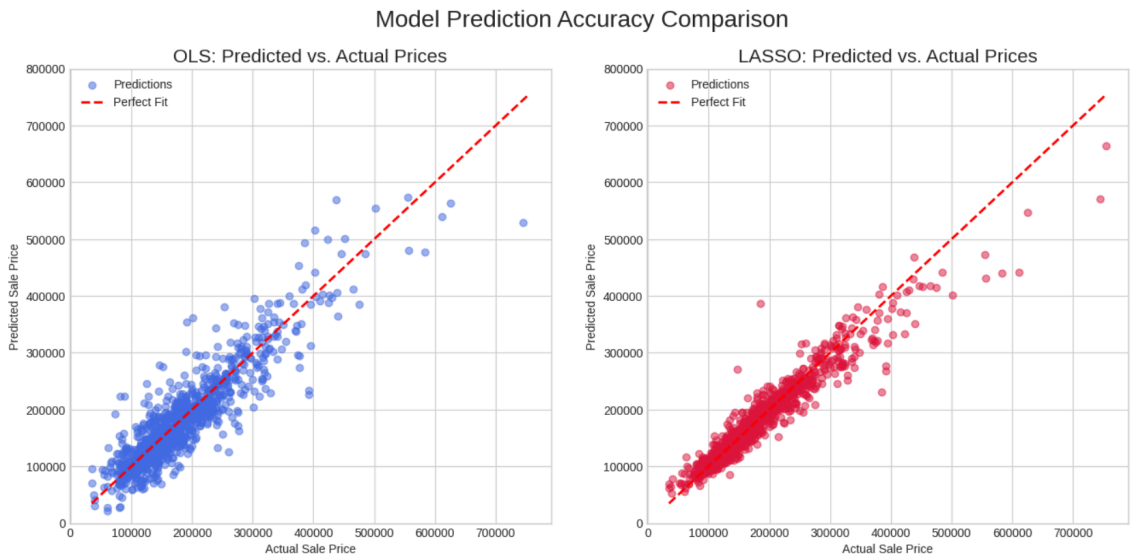
To form our LASSO Model, we performed cross-validation to select the optimal Alpha (λ) value which determines the severity of the penalty that LASSO applies to more complex models. The red dashed line on the graph below indicates the optimal Alpha value found through cross-validation, which is the value of Alpha that resulted in the lowest average MSE across the different folds of the cross-validation. This optimal Alpha value is then used in the final LASSO model to balance fitting the training data and preventing overfitting.

Results

Model Evaluations

To evaluate our models, we used two key performance metrics: Root Mean Squared Error (RMSE) and R-Squared (R2). The RMSE metric measures the average magnitude of the errors between the predicted and actual sale prices. A lower RMSE indicates better model performance, as the predictions are closer to the true values. The R2 metric represents the proportion of the variance in the dependent variable (SalePrice) that is predictable from the independent variables. An R2 score closer to 1 indicates that the model explains a larger portion of the variance and fits the data better.

Metric	OLS Model	LassoCV Model
RMSE	80874.879790	29995.832878
R2 Score	0.673081	0.905634



Based on the performance metrics and predictions, the LassoCV Model significantly outperformed the OLS Model. It achieved a much lower RMSE (29995.83 vs. 80874.88), indicating that its predictions were, on average, much closer to the actual sale prices. This distinction can be seen in the Predicted vs. Actual sales price graphs where LASSO predictions are more tightly grouped around the perfect fit line.

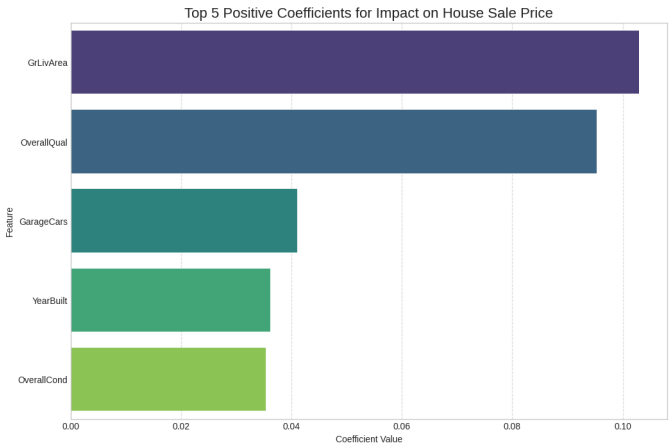
The LassoCV Model also had a substantially higher R2 score (0.906 vs. 0.673), meaning it explained a much larger proportion of the variance in the sale prices compared to the OLS model. This performance is likely due to LASSO's regularization, which helped it prevent overfitting to the training data by selecting the most relevant features rather than arbitrarily including all 233 features like the OLS model.

Although some prediction outliers were observed in the LASSO model's test results, it demonstrated significantly greater average effectiveness than the baseline OLS model. This predictive accuracy renders the LASSO model suitable for use in guiding business decisions.

Conclusion

In this analysis, our goal was to predict house sale prices using a comprehensive dataset of house and property features. We explored the data, handled missing values, and preprocessed the features, including one-hot encoding categorical variables and scaling numerical features. We compared two linear regression models: OLS as a baseline and LASSO regression that automatically selected key features. Our evaluation using RMSE and R2 as key metrics showed that the LASSO model significantly outperformed the OLS model.

Our LASSO model also identified the most impactful features for predicting house prices. The following features were associated with higher sale prices and had the most positive coefficients: GrLivArea (above ground living area), OverallQual (overall finish and material quality), GarageCars (car capacity of garage), YearBuilt (original construction date), OverallCond (overall condition rating). The "better" these features are (for example, the larger square footage of living area or the newer the house) the higher predicted sale price a house will have.



Practical Applications

In the context of our business problem, these findings are important to determining house prices in Ames because we are able to estimate house prices before we put them on the market. This will allow us to have more accurate prices and therefore sell houses quicker by matching the market willingness to pay. Our findings can inform our pricing strategy so that our real estate agents can focus marketing on the features most valued by the market.

Future Analysis and Next Steps

1) **Engineer New Features:** In future analysis, we can potentially engineer new features, such as interaction features between highly impactful variables to identify synergy between multiple variables. We can also create variables such as the age of the house (YrSold-YearBuilt) or the time since the last remodel (YrSold-YearRemodAdd) to capture these factors in one feature.
2) **Leverage More Advanced Models:** We could explore methods like Random Forests or Gradient Boosting Machines to capture non-linear relationships and interactions.
3) **Incorporating Additional Data:** We could gather external data sources, such as local park system scores, walkability scores, proximity to amenities, and school district ratings in Ames to further enhance our model and existing features such as where the house is located.